

CH 1 — Abstract Vector Spaces

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Ideas: $\mathbb{R}^n$ is useful. There are other objects that “behave” and “look” like $\mathbb{R}^n$

Examples:

1. $V = P_b(\mathbb{F}) = \{a_0 + a_1x + a_2x^2 + \dots + a_nx^n : a_i \in \mathbb{F}\}$

Polynomial degree $\leq n$:

Additive property

$$a_0 + a_1x + \dots + a_nx^n + b_0 + b_1x + \dots + b_nx^n = (a_0 + b_0) + (a_1 + b_1)x + \dots + (a_n + b_n)x^n$$

Scalar Multiplicative property

$$c(a_0 + a_1x + \dots + a_nx^n) = ca_0 + ca_1x + \dots + ca_nx^n$$

2. $V = M_{m \times n}(\mathbb{F}) = \left\{ \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \dots & \dots & \dots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} : a_{ij} \in \mathbb{F} \right\}$

Same properties for matrices like the above

Observation:

$\mathbb{R}^4$	$P_3(\mathbb{R})$	$M_{2 \times 2}(\mathbb{R})$
$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$	$a + bx + cx^2 + dx^3$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Notice that the above not only “look alike”, they share similar operations (addition and scalar multiplication)

This is isomorphism

3. $V = F(\mathbb{R}, \mathbb{R}) = \{\text{functions}, f : \mathbb{R} \rightarrow \mathbb{R}\}$

We can add and scalar multiply

Question: Does $F(\mathbb{R}, \mathbb{R})$ is isomorphic to $\mathbb{R}^n$

ANS: No, the reason will be learned later in the course. The current explanation is that $F(\mathbb{R}, \mathbb{R})$ is an infinity-dimensional vector space.

Definition – Vector Space

A vector space over $\mathbb{F}$ is a set V together with two operations $\begin{cases} \text{addition}(+) \\ \text{scalar multiplication}(\cdot) \end{cases}$ s.t.

$$\forall \vec{x}, \vec{y}, \vec{x} + \vec{y} \subseteq V$$

$$\forall c \in \mathbb{F}, c\vec{x} \subseteq V$$

Furthermore, the following conditions must hold for $\forall \vec{x}, \vec{y}, \vec{z} \in V, c, t \in \mathbb{F}$

1. $\vec{x} + \vec{y} = \vec{y} + \vec{x}$
2. $(\vec{x} + \vec{y}) + \vec{z} = \vec{x} + (\vec{y} + \vec{z})$
3. $c(\vec{x} + \vec{y}) = c\vec{x} + c\vec{y}$
4. $c(t\vec{x}) = ct(\vec{x})$
5. $(s + t) \cdot \vec{x} = s \cdot \vec{x} + t \cdot \vec{x}$
6. $1 \cdot \vec{x} = \vec{x}$
7. $\exists \vec{0} \in V$ s.t. $\vec{x} + \vec{0} = \vec{x}$
8. $\forall \vec{x} \in V, \exists -\vec{x} \in V$ s.t. $\vec{x} + (-\vec{x}) = \vec{0}$

Warning – Zero Vector and Additive Inverse

Zero vector $\vec{0}$ is not the vector with every entry being 0. It is subject to change with its corresponding vector space

Additive Inverse $-\vec{x}$ are not the vector with every entry being the negation of its origin entry. Refer to the Axiom to determine the corresponding additive inverse

Both $\vec{0}$ and $-\vec{x}$ are unique in their corresponding vector space

Note:

1. The elements in V are called vectors
2. We omit the “.” in $s \cdot \vec{x}$ to $s\vec{x}$
3. We write $\vec{x} - \vec{y}$ instead of $\vec{x} + (-\vec{y})$
4. $\vec{0}$ is called the **zero vector**
5. $-\vec{x}$ is called the **additive inverse** of $\vec{x}$

Examples:

$$1. V = \mathbb{F}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} : x_i \in \mathbb{F} \right\}, V = M_{m \times n}(\mathbb{F}) = \left\{ \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} : a_{ij} \in \mathbb{F} \right\},$$

and $V = P_n(\mathbb{F}) = \{a_0 + a_1x + \dots + a_nx^n : a_i \in \mathbb{F}\}$ with addition and scalar multiplication be usual operations.

$$\text{Here, } \vec{0} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix}, 0 + 0x + \dots + 0x^n$$

2. $V = F(\mathbb{R}, \mathbb{R}) = \{\text{functions}, f : \mathbb{R} \rightarrow \mathbb{R}\}$ with addition and scalar multiplication be usual operations.

Here, $\vec{0}$ = the zero function, i.e. $z(x) = 0$

3. $V = \mathbb{R}^2$

Define addition and scalar multiplication by

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 + y_2 \\ x_2 + y_1 \end{bmatrix}$$

$$s \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} sx_2 \\ sx_1 \end{bmatrix}$$

This is NOT a vector space.

$$1\vec{x} \neq \vec{x} \implies \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \neq \begin{bmatrix} x_2 \\ x_1 \end{bmatrix}$$

$$\vec{x} + \vec{y} \neq \vec{y} + \vec{x}$$

4. $V = \mathbb{R}^2$

Define addition and scalar multiplication by

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 + y_1 - 1 \\ x_2 + y_2 + 1 \end{bmatrix}$$

$$s \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} sx_1 + s - 1 \\ sx_2 - s + 1 \end{bmatrix}$$

Check axiom 7 and 8 as 1-6 are easy and straightforward

Axiom 7:

Note that in this case $\begin{bmatrix} 0 \\ \dots \\ 0 \end{bmatrix}$ is not the zero vector for this vector space!

We suppose $\begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$,

$$\text{then } \vec{x} + \vec{0} = \vec{x} \iff \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \iff \begin{bmatrix} x_1 + z_1 - 1 \\ x_2 + z_2 + 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \implies \vec{z} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \in V$$

Axiom 8:

We suppose $\begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$

$$\text{then } \vec{x} + \vec{z} = \vec{0} \iff \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \iff \begin{bmatrix} x_1 + z_1 - 1 \\ x_2 + z_2 + 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \implies \vec{z} = \begin{bmatrix} x_1 + 2 \\ x_2 - 2 \end{bmatrix}$$

5. Proof uniqueness of $\vec{0}$

Let V be a vector space. Suppose $\vec{0}_1, \vec{0}_2 \in V$ satisfy

$$\vec{0}_1 + \vec{x} = \vec{x}, \vec{x} \in V$$

$$\vec{0}_2 + \vec{x} = \vec{x}, \vec{x} \in V$$

We want $\vec{0}_1 = \vec{0}_2$

Proof